**4.**

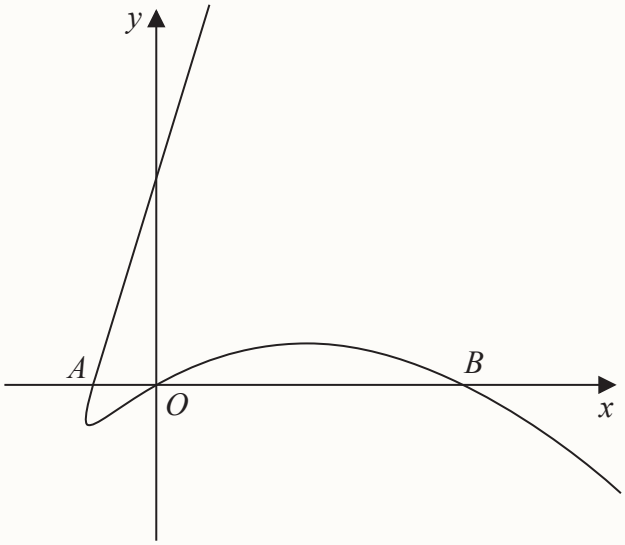



**Figure 2**

Figure 2 shows a sketch of part of the curve with parametric equations

$$x = 2t^2 - 6t, \qquad y = t^3 - 4t, \qquad t \in \mathbb{R}$$

The curve cuts the $x$-axis at the origin and at the points $A$ and $B$, as shown in Figure 2.

(a) Find the coordinates of $A$ and show that $B$ has coordinates (20, 0).

**(3)**

(b) Show that the equation of the tangent to the curve at $B$ is

$$7y + 4x - 80 = 0$$

**(5)**

The tangent to the curve at $B$ cuts the curve again at the point $P$.

(c) Find, using algebra, the $x$ coordinate of $P$.

**(4)**